



PYTHON • SQL • POWER BI

# Customer Sales Behaviors Analysis

Analyzing 3,900 transactions across product categories to uncover spending patterns, customer segments, and subscription insights.

DATASET OVERVIEW

# Dataset Summary

3,900

Total Rows

Transactions analyzed

18

Columns

Features per record

37

Missing Values

In Review Rating column

Key features span customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season), and shopping behavior (discount, promo codes, review ratings, shipping type).

## DATA PREPARATION

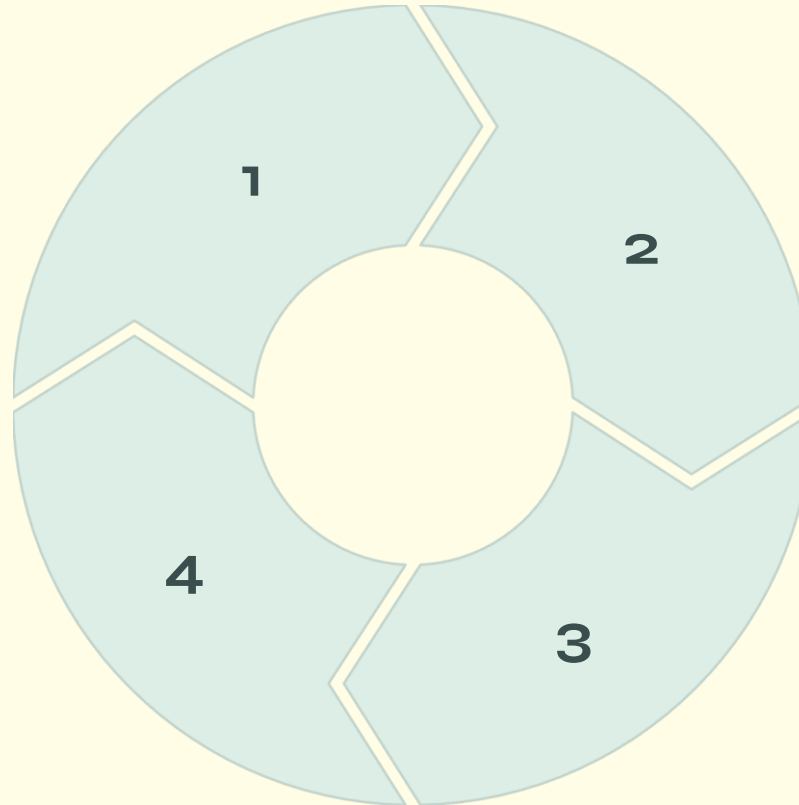
# Cleaning & Preprocessing in Python

### Explore Structure

Use `head()` and `info()` to inspect.

### Load Data

Read CSVs into pandas DataFrames.



### Fill Missing Values

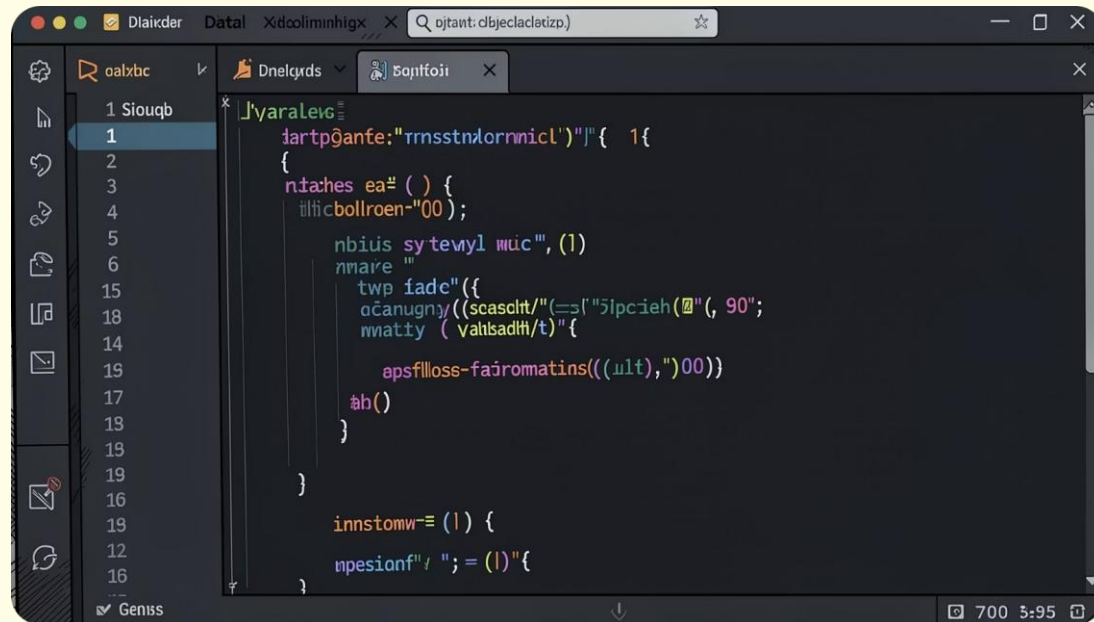
Impute ratings by category median.

### Clean Columns

Standardize names and drop duplicates.

Missing review ratings were filled using the median of each product category. Column names were standardized to lowercase with underscores, and the duplicate `promo_code` column was dropped since it matched `discount_applied`.

# Creating New Analytical Features



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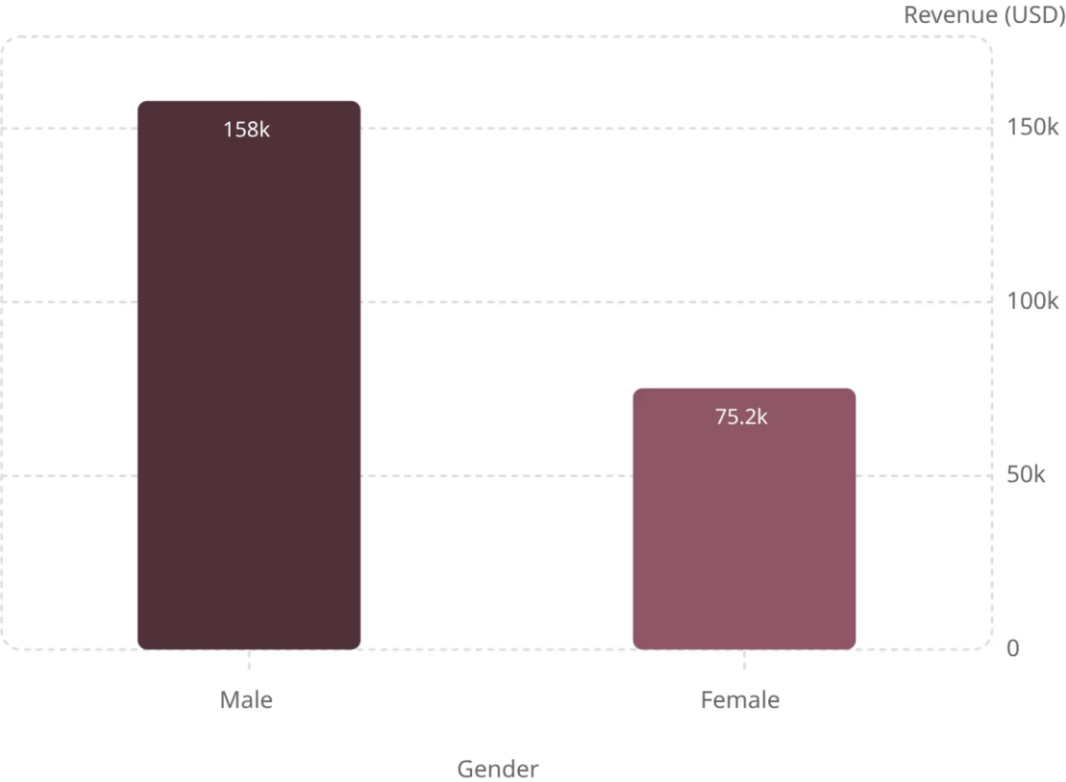
## Age Grouping

Customers categorized into young\_adults (18–25), adults (25–40), middle-aged (40–60), and seniors (60+).

## Frequency Conversion

Purchase frequency mapped to numeric days: Weekly = 7, Fortnightly = 14, Monthly = 30, Quarterly = 90, Annually = 365.

# Revenue & Customer Segmentation



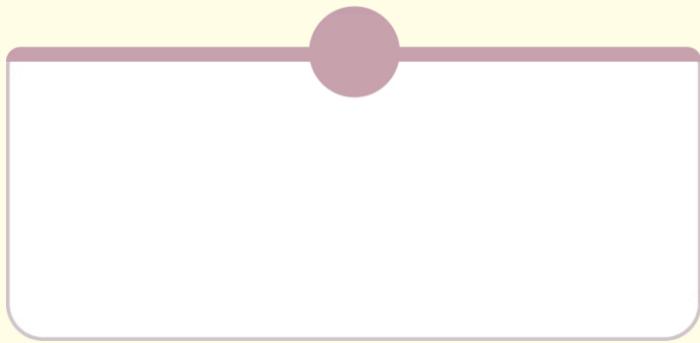
## Revenue by Gender

Male customers generated \$157,890 – more than double the \$75,191 from female customers.

## Customer Segments

Loyal: 3,116 · Returning: 701 · New: 83

# Top Products & Shipping Performance



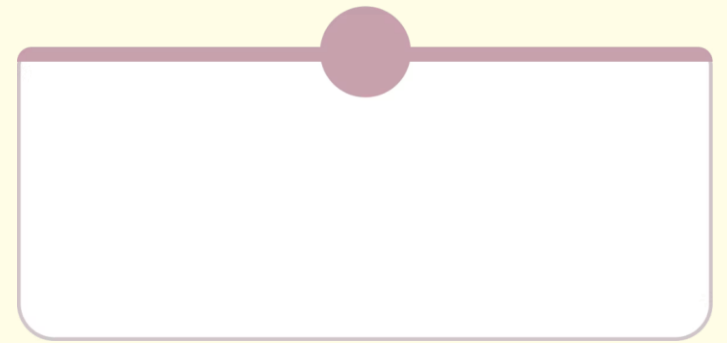
## Highest Rated Items

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)



## Highest Discount Rates

Hat (50%), Sneakers (49%), Coat (49%), Sweater (48%), Pants (47%)



## Shipping Revenue

Express averages \$60.48 vs. Standard at \$58.46 per purchase

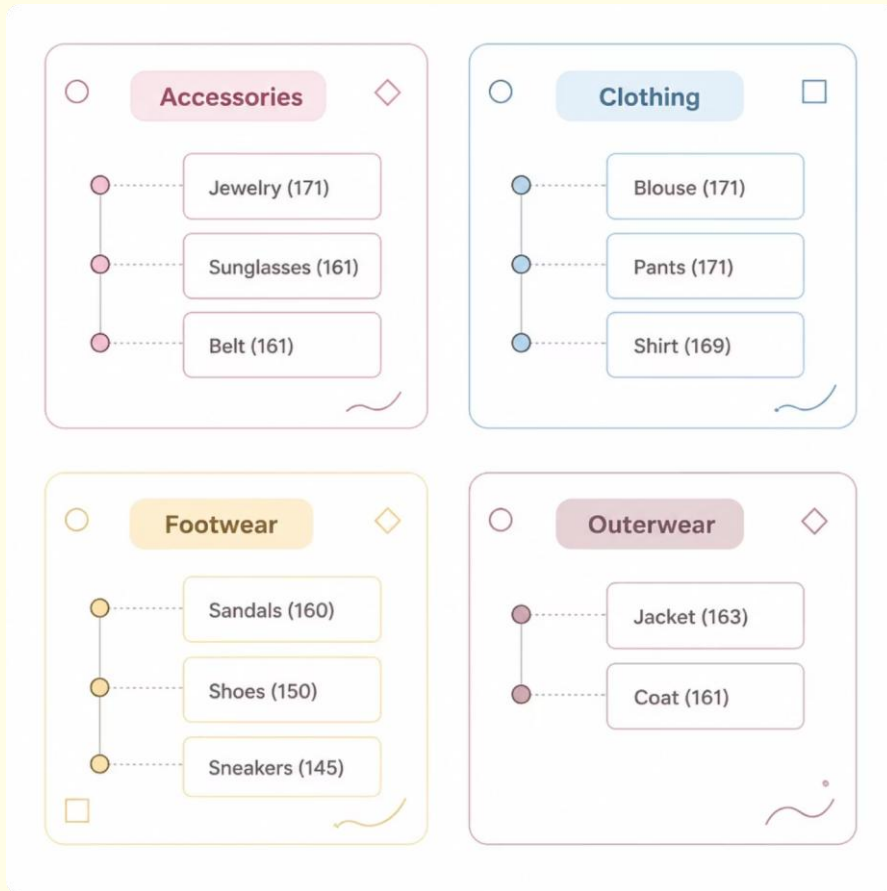
## SQL INSIGHTS

# Subscription & Category Breakdown

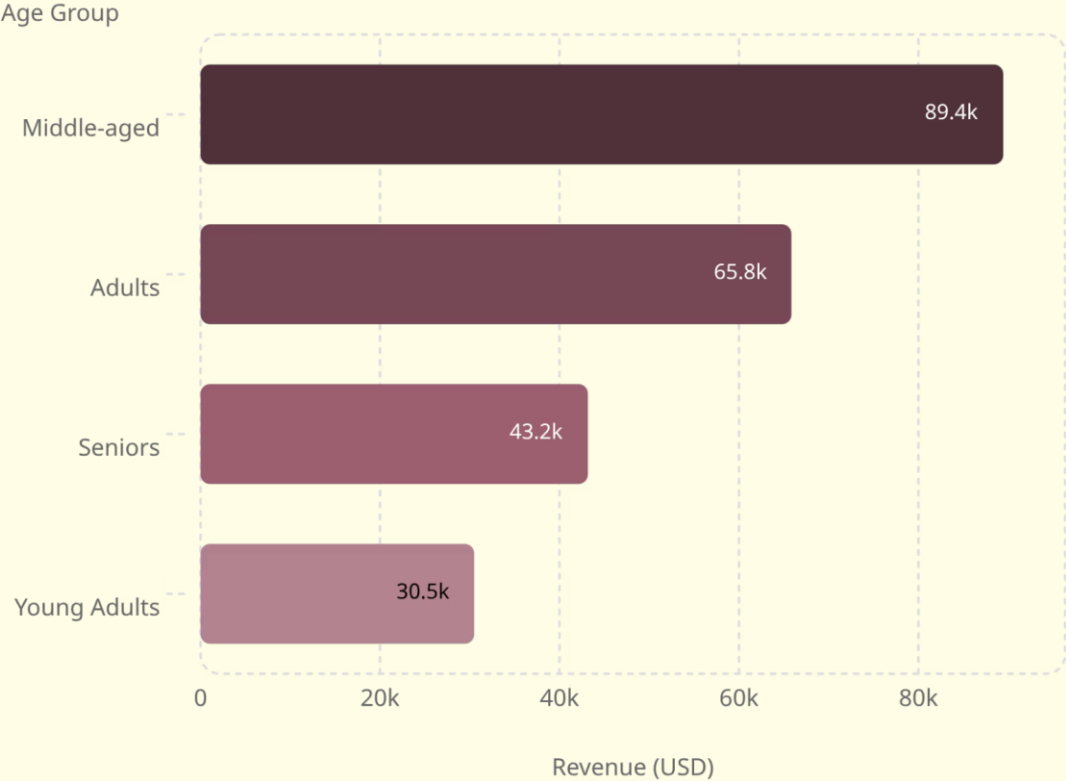
## Subscription Impact

Non-subscribers: 2,847 customers, \$170,436 revenue (\$59.87 avg). Subscribers: 1,053 customers, \$62,645 revenue (\$59.49 avg).

Repeat buyers are mostly non-subscribers (2,518) vs. subscribers (958).



# Revenue Contribution by Age Group



## Key Findings

Middle-aged customers drive the highest revenue at \$89,445, followed by adults at \$65,842. Seniors contribute \$43,164, while young adults account for \$30,491.



# Customers Sales Dashboard

An interactive Power BI dashboard combining KPIs, charts, and slicers for dynamic analysis of customer behavior, revenue, and sales performance.

3.9K Customers

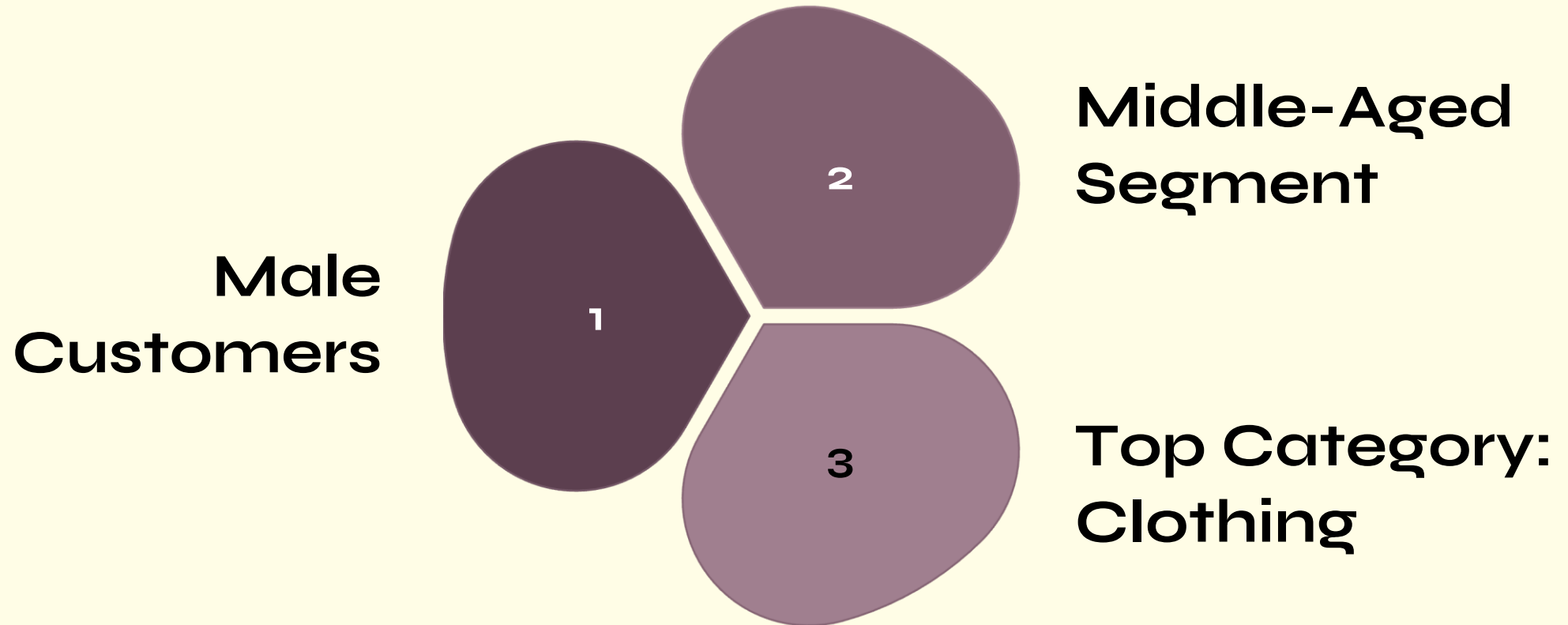
\$59.76 Avg Purchase

3.75 Avg Review



## SUMMARY

# Key Takeaways



The dashboard enables data-driven decisions by combining interactive filters for gender, payment method, subscription status, discounts, and product category – revealing where revenue concentrates and which segments to target.